

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 223 — Past Questions & Materials
by Renance

Course: MTH 223
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — MTH 223 QUESTIONS AND ANSWERS

Level 200 • Second Semester • PastQuestion

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(69)

Solution

We have $y = ax^n$

Taking logarithms, $\log_{10} y = \log_{10} a + n \log_{10} x$ — (i)

ie $Y = A + nX$

Where $X = \log_{10} x$, $Y = \log_{10} y$, $A = \log_{10} a$.

Divide the data into two groups each containing four pairs of values, so that

x	y	$X = \log_{10} x$	$Y = \log_{10} y$
10	1.06	1.0253	0.0253
20	1.33	1.3010	0.1238
30	1.52	1.4771	0.1878
40	1.68	1.6021	0.2253
		$\Sigma X = 5.4055$	$\Sigma Y = 0.5562$

x	y	$X = \log_{10} x$	$Y = \log_{10} y$
50	1.71	1.6990	0.2577
60	1.91	1.7782	0.2810
70	2.01	1.8451	0.3032
80	2.11	1.9031	0.3243
		$\Sigma X = 7.2254$	$\Sigma Y = 1.1662$

Substituting the average of X 's and Y 's of the two groups in (i) we get

$$\frac{0.5562}{4} = A + n \frac{5.4055}{4} \quad \text{ie} \quad 0.1390 = A + 1.3514n$$

(ii) - (ii) gwc

(70)

$$0.1526 = 0.455n$$

$$\text{i.e. } n = 0.3354$$

$$\text{And from (ii) } A = -0.3142$$

$$\text{i.e. } \log_{10} A = -0.3142$$

Whence

$$A = \text{antilog}(-0.3142) = 0.4851$$

= Other topics are

- * Principle of least square
- * Method of least square
- * Method of Moments (by integration)

(75)

(71) Application to Curve fitting, Circuit and Solve Engineering Problems

Now, we use this method [Gauss-Jordan] to solve problems which reduces to linear equations because there are some problems that need themselves to a linear algebra.

Let us consider the standard example

Exple. Find a polynomial $p(x)$ whose graph $y=p(x)$ fits through the points $(0, -2.7)$, $(2, -4.5)$ and $(1, 0.97)$, with a quadratic polynomial $p(x) = Ax^2 + Bx + C$ where A , B and C are the coefficients of the independent variable

Solution

$$At \quad p(0) = C = -2.7$$

$$p(2) = 4A + 2B + C = -4.5 \Rightarrow$$

$$p(1) = A + B + C = 0.97$$

$$\left(\begin{array}{ccc|c} A & B & C & \\ \hline 0 & 0 & 1 & -2.7 \\ 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 0.97 \end{array} \right)$$

By Gauss-Jordan rref Method

$$ie \text{ rref} \left(\begin{array}{ccc|c} 0 & 0 & 1 & -2.7 \\ 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 0.97 \end{array} \right) = \left(\begin{array}{ccc|c} 1 & 0 & 0 & -4.52 \\ 0 & 1 & 0 & 8.14 \\ 0 & 0 & 1 & -2.7 \end{array} \right)$$

$$\text{Therefore } A = -4.52$$

$$B = 8.14$$

$$C = -2.7$$

Exple find which polynomial $Q(x)$ have graph $y=Q(x)$ which fits through the point $(0, -2.7)$, $(2, -4.5)$ and $(1, 0.97)$. Let $Q(x) = Ax^3 + Bx^2 + Cx + D$.

soln

(72)

$$Q(0) = \Delta = -2.7$$

$$Q(2) = 8A + 4B + 2C + \Delta = -4.5$$

$$Q(1) = A + B + C + \Delta = 0.97$$

A	B	C	Δ	
0	0	0	1	-2.7
8	4	2	1	-4.5
1	1	1	1	0.97

By rref(A) we have

$$\text{rref} \left(\begin{array}{cccc|c} 0 & 0 & 0 & 1 & -2.7 \\ 8 & 4 & 2 & 1 & -4.5 \\ 1 & 1 & 1 & 1 & 0.97 \end{array} \right) = \left(\begin{array}{cccc|c} 1 & 0 & -0.5 & 0 & -4.07 \\ 0 & 1 & 1.5 & 0 & 7.69 \\ 0 & 0 & 0 & 1 & -2.7 \end{array} \right)$$

Therefore

$$A = -4.07 + 0.5C$$

$$B = 7.69 - 1.5C$$

$$C = C$$

$$\Delta = -2.7$$

$$\text{There } Q_c(x) = (-4.07 + 0.5C)x^3 + (7.69 - 1.5C)x^2 + Cx - 2.7$$

(79)

AESA ^{CSC} ~~4/6~~ 4/6

1. Round off the numbers 72525012 and 0.00324 to 5 significant figures and compute $\ln$ and $\exp$. ✓
2. Solve the equation $3x^3 - 4x^2 + x + 88 = 0$ with one of its roots being $2 + \sqrt{7}i$.
3. Find the larger root of $x^2 - 2x - 3.5 = 0$ using Newton-Raphson method. ✓
4. If $\Delta f_n = f_{n+2} - f_{n+1}$, evaluate $\Delta^3 f_n$. ✓
5. Find an approximate root of $x^3 - x - 2 = 0$ in the interval $[1, 2]$ with a Tol 10^{-5} accuracy, using the Newton's Raphson method.
6. Find a real root of the equation $x^3 - 2x - 5 = 0$ by the method of ~~false~~ Regular false method, correct to 3dp with a root lie between 2 and 3. ✓
7. Construct a difference table with the function $f(x) = x^4 - 2x + 1$ at $0(0.4)$ ✓

Calculus, Gen
Define - Calculus,
Find the $\lim_{x \rightarrow 4}$
Define gradient
Differentiate f
And $y = 2x^2$
If $y = 16x^3$
Differentiate
 $y =$
 $y =$
If
If
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=

Calculus, Geometry, Linear Algebra (2) MAT 1105

Define - Calculus, And Differential Calculus

Find the $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$, $\lim_{x \rightarrow \frac{1}{2}} (x^4 - 5x^3 - 2x^2 - x + 1)$

Define Gradient.

Differentiate from first principle $y = \sqrt{x}$ w.r.t x

And $y = 2x^3 + 5x + 1$ w.r.t x

If $y = 16x^3 + \frac{1}{\sqrt{x}}$ differentiate w.r.t x

Differentiate the following

$y = 6x^3 \sin x \cos x$

$y = \frac{2x^2}{\sqrt{3x^2 - 2x + 1}}$

If $y =$ Define Explicit, And Implicit function

If $y^3 x^2 + 2x^2 y + 4x^2 y^4 = 2x \sin x$

Find $\frac{dy}{dx}$ at $(x, y) (0.5, 1)$

Differentiate $y = \frac{(x^3 + 2x - 1)^4 (3x^2 + 2)^3}{(2x^2 + 2x + 1)^3}$ w.r.t x

Differentiate $y = \sin^{-1} \frac{x}{a}$ w.r.t x

If $f(x) = \sin 2x$ show that $f''(x) + f'(x) + 4f(x) - 2 \cos 2x = 0$

Find the Min and the Max of the fcn $y = 2x^3 - 9x^2 + 12x + 4$

(3)

- One sketch the graph
- = Define Integration
- = Evaluate $\int (\sqrt{x} - \frac{1}{\sqrt{x}}) dx$
- = Evaluate $\int (5x^2 - 3) dx$
- = Evaluate $\int \frac{7x-2}{2x^2+11x+5} dx$
- = Find the distance between the points (3, -2) and (-4, 5)
- = ~~Find~~ If P and Q divide the line AB, such that point A(-1, 2) and B(3, 4) internally and externally in the ratio 5:3. Find their coordinates
- = Find the gradient of the line joining (3, -2) and (-5, 8) and the angle it makes with the x-axis.
- = A straight line has a gradient of $-\frac{3}{2}$ and passes through the point (1, 4). Find its gradient and the intercept on the y-axis.
- = Find the equation of a line AB, which passes through the points (1, -1) and (-2, -13). Hence find the equation of the straight line CD parallel to AB, which passes through the point (2, 11)
- = Define Matrix, Row and Column of a matrix.
When is a matrix said to be square matrix
- = If $A = \begin{pmatrix} 3 & 2 & 4 \\ -2 & 4 & 3 \\ 0 & - & 2 \end{pmatrix}$. Find the determinant of A

Define Numerical Analysis, forms
 the Art of Numerical Analysis
 the Round off of the Number 8652521 and 3724265
 to 4 s.f. Compute the ϵ_a , ϵ_r , ϵ_p in each case

If $\frac{121.31}{10.982}$ find the Max relative error.

If $\frac{(2.136)^2 - (1.184)^2}{(2.136)^2 - (1.179)^2}$ find the Max absolute and relative error.

State the basic properties of Equations
 Have if $f(x) = 2x^3 + x^2 - 13x + 8 = 0$. Find all the factors.
 Solve the equation $3x^3 - 7x^2 + x + 8 = 0$ with one of the roots being $2 + \sqrt{2}i$

Consider
 $4x_1 + x_2 - x_3 = 4$ — (1)
 $2x_1 + 3x_2 + x_3 = 4$ — (2)
 $x_1 + x_2 + 5x_3 = 18$ — (3)

Solve by Gauss-Jordan method.

Given
 $10x_1 + x_2 + 3x_3 - x_4 = 2$
 $x_1 - 10x_2 + 5x_3 - x_4 = -2$
 $2x_1 + 2x_2 - 20x_3 - 2x_4 = -1$
 $x_1 + x_2 + 3x_3 - 10x_4 = 3$

Use Jacobi, solve the unknown variable x_1, x_2, x_3 and x_4

Given
 $10x_1 + x_2 + 3x_3 - x_4 = 2$
 $x_1 - 10x_2 + 5x_3 - x_4 = -2$
 $2x_1 + 2x_2 - 20x_3 - 2x_4 = -1$
 $x_1 + x_2 + 3x_3 - 10x_4 = 3$

Use Gauss-Seidel, find the unknown variable x_1, x_2, x_3 and x_4

- = By Graphical method, find all the possible closed interval of this non-linear function $f(x) = 8x^3 - 12x^2 - 2x + 3$ where $-4 \leq x \leq 4$.
 $f(x) = 3 - x = e^{x-1}$
- = Determine approximately how many iterations are necessary to solve $f(x) = x^3 - 25$, with an accuracy of $\epsilon = 10^{-4}$ within an interval of $[2, 3]$. Hence show that $f(x)$ has at least one zero in the interval $[2, 3]$, using Bolzano method.
- = Find the roots of the equation $xe^x = \cos x$ using regular false method correct to four decimal places.
- = Find by Newton's Raphson method $3x = \cos x - 1$ at an interval $[0, 1]$, with a TA 10^{-5} .
- = Given $f(x) = 4x^3 - 2x - 5$ Construct a Difference table at $O(2)^{12}$.
- = If $\Delta_{f_n}^2 f_{n+1} - f_n$ find $\Delta_{f_n}^9$.
- = Find the factors of the equation $x^3 + 6x^2 + 11x + 6 = 0$.
- = Define, Iteration
- = State the theorem that satisfies Bolzano Iterative method.
- = Find a root of the equation $x^3 - 4x - 9 = 0$, using Bolzano Iterative method in four stages with a close interval $[2, 3]$. Also predict the total number of iterations.

Paper 2 of 3 — MTH 223 2022/2023

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MERICAL ANA\ elie '3C). SePtember, 2028. oration: apo

Jer graeles One (1) and any of the (3) questions

all be given for brevity andcoi eee

Se answers:

question 1

(a) when is & function $f(x, y)$ sald to belli

Uf ind the approximate root off) =

with 10^{-7} lever of.accuracy, Using the Newton = Raphson S Method.

(c) Find the minimumroot of $f(x) =$

schitzian.

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~Qin the inter val(y 2]

1

(95 Marks)

3 $12x^? + 47x \sim 60 = 0$, Using

the Srinivasa Ramanujan's Method(stopping at $n = 5$), 3+ marks

p (435 marks)

Question 2 7

Given the following data points;

I that interpolates this datausing:

polation formitla, (72 marks)

if ference interpolation formula (7 marks)

P. Examine the followlig

(24 marks)

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Paper 3 of 3 — MTH 223 PQ 2024

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
DEPARTMENT OF MATHEMATICS
B.Sc. DEGREE EXAMINATION (2023/2024), SECOND SEMESTER, MTH 223
(NUMERICAL ANALYSIS I 3C), October, 2024. Duration: 2½ Hours.

Instructions: Answer question one (1) and any other three (3) questions.
Credit shall be given for brevity and concise answers.

Question 1

(a) When is a function $f(x, y) \forall x, y \in D$ said to be Lipschitzian? (1 mark)

(b) Let x and x^* represent exact and approximate values respectively. Define Absolute and Relative error in terms of x and x^* . (2 marks)

(c) Solve the IVP $y' = xy$, $y(0) = 1$, $h = 0.1$, using the 4th order Runge-kutta method and stopping at $n = 9$. (22 marks)

Question 2

(a) Construct the forward difference table corresponding to the function $f(x) = 1 - x^2$, $x = 1, \dots, 4$. (3 marks)

(b) Find the minimum root of the equation $x^3 - 18x^2 + 107x - 210 = 0$, using the Srinivasa Ramanujan's method (stopping at $n = 5$). (12 marks)

Question 3

Find the interpolating polynomial for the table

x	4	2	1
$f(x)$	8	3	5

Using

(a) Lagrange interpolating polynomials (LIP) (7.5 marks)

(b) Newton's divided difference formula (NDDF) (7.5 marks)

Question 4

(a) Given that $f(x) = 1 - x$ and $g(x) = x - 1$, evaluate $\Delta f(x) - \nabla g(x)$. (5 marks)

(b) Find the approximate root of $f(x) = x^2 - 100$ in the interval $[9, 11]$ with 10^{-7} level of accuracy, using the Newton-Raphson's method. (10 marks)

Question 5

(a) Compute $f''(2.1)$ from the following tabular data (8 marks)

x	1.5	1.7	1.9	2.0	2.1
$f(x)$	3.0551	3.9529	5.0495	6.3980	8.0249

(b) Evaluate $I = \int_0^1 xe^x dx$, using the series expansion method. (7 marks)

Question 6

(a) Using the predictor-corrector method, solve the IVP $y' = x - y$, $y(0) = 1$ with $h = 0.01$ (stopping at $n = 2$). (8 marks)

(b) Using Taylor's series method, find y at $x = 0.4$ if $y' = 1 - xy$, $y(0) = 2$. (7 marks)

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